

# HARSHA KARTHIKEYAN

## SOFTWARE ENGINEER

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### PROFESSIONAL SUMMARY

**.NET Developer** with **4 years** of experience building and supporting healthcare SaaS platforms, most recently a notification platform processing 30M+ notifications a year. Specialist in **C#/.NET 8, ASP.NET Core, SQL Server**, designing **REST APIs, microservices and event-driven systems on Azure**, with working experience in **Angular**. Stepped up to **lead technical delivery for two Lines of Business**, delivering every release on its planned date.

### TECHNICAL SKILLS

**Languages:** C#, SQL, TypeScript, JavaScript

**Frameworks:** .NET 8, ASP.NET Core, Entity Framework Core, Web API, Angular, LINQ

**Cloud & DevOps:** Microsoft Azure (App Services, Web Jobs, File Storage, Application Insights), Azure DevOps CI/CD

**Database & Messaging:** Microsoft SQL Server, Kafka

**Development Practices & Tools:** Agile, Scrum, Git, GitHub Copilot, Postman, Swagger, JIRA

**Architecture:** OOP, SOLID Principles, REST APIs, Microservices, Multi-tenancy, Dependency Injection, JWT, Caching

**Testing:** Unit Testing, TDD, xUnit

### PROFESSIONAL EXPERIENCE

#### Optum

Jul 2022 - Jun 2026

*Primary Project: Notification Generation System (NGS): a healthcare SaaS platform automating member communications in a regulated environment with audit requirements, processing 30M+ notifications a year across 10+ Lines of Business.*

#### Software Engineer

Mar 2024 - Jun 2026

- **Led technical delivery for two Lines of Business**, owning solution design, estimation and code review, working with US-based stakeholders, and delivering every release on its planned date.
- Onboarded both Lines of Business onto the platform, handling **250K+ notifications weekly** for **3,000+ concurrent users** and saving **~\$200K for the business**.
- Automated the manual intake process for **7K-10K letters a day**, saving 5-8 minutes of manual effort per letter.
- Migrated **40 Kafka topics across 4 environments to Confluent Cloud** (ACLs, certificate-based authentication) with **zero messages lost**.
- Delivered application changes for the **migration from Windows VMs to Azure**, consolidating 7 LOB configuration databases into a **single multi-tenant database**, reducing Azure infrastructure cost.
- Monitored applications and resolved production incidents using Application Insights.

#### Associate Software Engineer II

Jul 2023 - Mar 2024

- **Built ASP.NET Core REST APIs** in C#, EF Core and SQL Server using SOLID principles and Dependency Injection, and integrated them with **Angular** front-end components.
- Upgraded 36 background projects from **.NET 6 to .NET 8**, replacing unsupported libraries like iTextSharp.
- Onboarded a new Line of Business through configuration-driven enhancements and Kafka and API integrations.
- Wrote **xUnit** unit tests for every user story and supported integration testing.

#### Technology Development Program (TDP) II

Jul 2022 - Jul 2023

- *Appeals & Grievance Tracking System (Jul - Nov 2022):* Built REST APIs, integrated them with **Angular user interfaces**, and maintained **stored procedures**.
- *NGS (Nov 2022 - Jul 2023):* Developed API and SQL features, and **.NET background jobs** for letter processing.

### EDUCATION

Bachelor of Technology - Electronics and Instrumentation Engineering  
SASTRA University | First class with distinction

Jul 2018 - Jul 2022

### ADDITIONAL INFORMATION

**Certifications:** Microsoft AZ-900, Generative AI (AI Dojo), UnitedHealthcare Medicaid (all levels), AHIP AHM 250

**Award:** M&R Technology Impact Award (2023)